

Zixuan (Xuan) Yin

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Data Science student at UC San Diego with hands-on experience building across the full stack of modern data science, from statistical modeling and machine learning to generative AI and data engineering. Background spans computer vision systems, AI-powered applications, and production analytics infrastructure across research and industry settings. Skilled in Python, SQL, PyTorch, and the broader ML ecosystem with a strong foundation in translating complex data into clear, actionable insights for technical and non-technical audiences.

EDUCATION

Bachelor of Science, Data Science | University of California San Diego | Expected Dec 2027

PROFESSIONAL EXPERIENCE

Data Analytics Intern | Elara | May 2026 – Sep 2026

- Partnered with business stakeholders to define product analytics requirements and translate ambiguous needs into an actionable project plan, designing and deploying a generative AI-enabled event tracking schema in PostHog covering 25+ user events ahead of MVP launch
- Built, tested, debugged, and validated the solution across the full pipeline, using metrics and stakeholder feedback to evaluate impact and improve quality while communicating progress and tradeoffs clearly to technical and non-technical partners

Algorithm Developer Intern | Boundary Remote Subsurface Solutions | Jan 2025 – Sep 2025

- Developed reusable Python pipelines for amplitude extraction, phase unwrapping, and geospatial alignment across multi-temporal SAR datasets, applying statistical diagnostics to validate interferometric results and identify subsurface anomalies
- Documented workflow logic and communicated complex processing findings through structured reports to engineering stakeholders, translating technical outputs into actionable insights

Code Instructor | Code Ninjas | Nov 2023 – Aug 2026

- Instructed students in JavaScript, C#, Python, and Unity game development through game-based curriculum, building deep familiarity with gaming products, customer engagement mechanics, and interactive entertainment technology while contributing to a center ranked in the top 2.5% of 400+ global locations

RESEARCH EXPERIENCE

Research Assistant | Halicicoglu Data Science Institute | UC San Diego | Dec 2025 – Present

- Aggregated and merged multi-source carceral datasets in Python, applying data exploration, cleaning, and mining techniques to identify patterns and recurring data gaps across jurisdictions
- Developed data visualizations and design artifacts to surface disparities in mental health court outcomes, producing technical documentation and communicating findings to interdisciplinary research teams

Research Intern (ESCAPE Program) | University of New Mexico | Jun 2024 – Aug 2024

- Collected and processed hourly air quality data across multiple Albuquerque monitoring sites, using Python and SQL to integrate multi-parameter datasets and isolate temperature as a key driver of seasonal pollutant fluctuations
- Produced structured findings and visualizations for program leadership, translating complex technical outputs into clear insights

PROJECTS

Sauced.AI — Agentic Meal Prep Assistant | 2026 | [Link](#)

- Identified a real operational problem and designed an AI prototype using the Gemini LLM API and Browser Use agent framework, implementing automated workflows for web scraping, nutritional analysis, and personalized recipe ranking through natural language interaction
- Iterated based on user feedback and metrics, improving solution quality and documenting outcomes and next steps for continued development

Vision-Based Gesture Control System | 2026 | [Link](#)

- Designed and prototyped a machine learning solution for real-time gesture classification, building, testing, and deploying a ResNet18 CNN using PyTorch and OpenCV on 285K+ augmented images achieving 85%+ confidence thresholds across 6 gesture classes
- Iterated on model performance through debugging and end-user validation, documenting outcomes, future use cases, and recommendations for improvement

The Fenty Effect: Data-Driven Analysis of Inclusivity in Cosmetic Shade Ranges | 2025 | [Link](#)

- Analyzed U.S. foundation product data (2016–2025) across 104 brands to identify category trends, market opportunities, and competitive dynamics, developing recommendations to reveal persistent brand-level disparities
- Created interactive dashboards and scorecards to communicate analytical findings clearly, presenting project outcomes and data-driven recommendations to diverse audiences

SKILLS

Data/Analysis: Python (pandas, NumPy, matplotlib, seaborn, scikit-learn, statsmodels), R, SQL, MATLAB, Tableau, Power BI, Spark (PySpark)
Modeling/Methods: Statistical modeling, regression analysis, ETL/ELT pipelines, machine learning (TensorFlow, PyTorch, scikit-learn), deep learning, computer vision, LLM integration, agentic AI workflows

Tools: Git/GitHub, Docker, AWS, Azure, Linux/Bash, PostHog, OpenCV, Hugging Face Transformers, Databricks, C/C++, Excel, Snowflake

Languages: English (fluent), Chinese Mandarin (fluent)

CERTIFICATIONS & AWARDS

LLMOps – Duke University | Outstanding Research Poster Award (2023) – Irvine Valley College Research Symposium